

1) (a) Hypothesis: $n > 1$, n is integer, n is not prime

Conclusion: $2^n - 1$ is not prime.

when $n=6$, hypothesis is true, as $2^6 - 1 = 63$ is not prime so theorem is True at instance $n=6$.

(b) At instance $n=15$, hypothesis is True, $2^{15} - 1 = 32767$ is not prime as $32767 = 151 \times 217$. So theorem is True

(c) For $n=11$, hypothesis is False. The theorem tells us nothing.

2) Hypothesis: $b^2 > 4ac$

Conclusion: $ax^2 + bx + c = 0$ has exactly 2 real solutions

(b) An instance of theorem is where we assign particular values to hypothesis & check if conclusion follows from hypothesis.

(c) $5^2 > 4(2)(3) \Rightarrow 25 > 24$

So $2x^2 - 5x + 3 = 0$ ^{should} have exactly 2 solutions

$$2x^2 - 2x - 3x + 3 = 0$$

$$2x(x-1) - 3(x-1) = 0$$

$$2x - 3 = 0 \text{ \& } x - 1 = 0$$

$x = 3/2$ & $x = 1$ are exactly 2 solutions. This instance tells us that theorem is correct.

(d) $4^2 = 16$ & $4ac = 4(2)(3) = 24$

as $b^2 < 4ac$, hypothesis is false & we can conclude nothing

3) Hypothesis: $n \in \mathbb{N}^+$, $n > 2$, n is not prime

Conclusion: $2n + 13$ is not prime

(b) $n = 3$ is a counter example $2(3) + 13 = 19$ is prime.

So for $n = 3$, hypothesis is T but conclusion is F.

4) Prove.

$$\frac{\text{Goal}}{a^2 < b^2}$$

$$0 < a < b$$

$$a \in \mathbb{R}^+, b \in \mathbb{R}^+$$

$$\text{As } a < b \quad a - b < 0 \text{ or } b - a > 0 \quad (0 < b - a)$$

multiplying by positive number $b + a$ on both sides as a and b are positive, $b + a$ is +ve.

$$\cancel{(a+b)} \cancel{(a-b)} \quad (b+a)(b-a) > (b+a) \times 0$$

$$b^2 - a^2 > 0 \rightarrow b^2 > a^2 \text{ or } a^2 < b^2$$

5) Prove: $a, b \in \mathbb{R}$, $a < b < 0$

$$\text{to: } a^2 > b^2$$

Assume $a < b$

Multiplying by a on both sides, since a is negative, inequality will be reversed

$$a^2 > ab$$

$$\text{Similarly for } b \quad ab > b^2$$

$$\text{So } a^2 > ab > b^2 \rightarrow a^2 > b^2$$

6) Prove: same as 4) $0 < a < b$

$$\text{Goal: } \frac{1}{b} < \frac{1}{a}$$

$$\text{as } a < b, \quad \frac{a}{a} = 1 < \frac{b}{a} \quad (\because \text{dividing by } b)$$

I am not writing my proof formally in this diary (in english statements)

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Again dividing by b we get $\frac{1}{b} < \frac{b}{a \times b} \rightarrow \frac{1}{b} < \frac{1}{a}$

7) Given $a^3 > a$, ask what $a^5 > a$

My approach: As a^2 is positive for every a . Multiplying by a^2 on both sides

$$a^5 > a^3$$

Also $a^3 > a$ so $a^5 > a^3 > a \rightarrow a^5 > a$

Book approach:

Key: Assume $a^3 > a$ (Note that this assumption already enforces $|a| > 1$ i.e. $a \neq 0, a > 1$ otherwise hypothesis will be F.)

$$\text{or } a^3 - a > 0$$

Multiply by $a^2 + 1$

$$(a^3 - a)(a^2 + 1) > 0 \quad (a^2 + 1) \text{ hypothesis is correct, the conclusion must also hold. If hypothesis is False, we cannot say anything about conclusion}$$

$$a^5 - a^3 + a^3 - a > 0$$

$$a^5 - a > 0$$

Notice that here $a^3 - a > 0$ by hypothesis (if we assume it's T) so for $(a^3 - a)(a^2 + 1) > 0$, $a^2 + 1 > 0$, ~~$a^3 - a > 0$~~

so cannot be between -1 & 1 (if $a = 0$, then $a^3 \neq a$)
so both min & max needs are true, both already enforce that $a^3 > a$ means $a \in [-1, 1]$ (Domain)

8) Given: $A \setminus B \subseteq C \cap D$

$$x \in A$$

$$x \notin D$$

Goal:

$$x \in B$$

This is naturally in goal of form $(p \rightarrow q)$ but we let p

in given & we have to prove q

if given $p \rightarrow q$ doesn't work so will use contrapositive

$p \rightarrow q$
 $\neg q \rightarrow \neg p$

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Given: $A \cap B \subseteq C \cap D$, $x \in A$, ~~$x \in B$~~ $x \notin B$

Goal: ~~$x \in B$~~ $x \in D$

$x \in A$ and $x \notin B$ means $x \in A \setminus B$

As $A \setminus B \subseteq C \cap D$, $\forall x (x \in A \setminus B \rightarrow x \in C \cap D)$

So $x \in C \cap D$ which means $x \in C \wedge x \in D$, so $x \in D$.

Therefore if $x \notin B$ then $x \in D$.

It's a bit difficult to prove it using $p \rightarrow q$
 lets us $A \cap B \subseteq C \cap D$

means $\forall x [x \in A \cap B \rightarrow x \in C \cap D]$

$\forall x [(x \in A \wedge x \in B) \rightarrow (x \in C \wedge x \in D)]$

$\therefore \forall x [\neg(x \in A \wedge x \in B) \vee (x \in C \wedge x \in D)]$

Assuming $x \in A$ & $x \notin D$ from hypothesis

$\forall x [\neg(T \wedge x \notin A) \vee (x \in C \wedge F)]$

$\forall x [\neg(x \notin B) \vee F]$

$\forall x (x \in B)$

Proved (For every x in domain,
 all hypothesis is imply the
 conclusion) as conclusion
 follows from all hypothesis

(Note that $\neg q \rightarrow \neg p$ method was easier)

9) Given: $A \cap B \subseteq C \cap D$, $x \in A$, $x \in D$
 Conclude $x \in B$

By contrapositive, if in any proof we will prove if $x \in B$
 then $x \in D$

Assuming $x \in B$, $x \in A$, $x \in A \cap B$. $A \cap B \subseteq C \cap D$

so $x \in C \cap D$ which means $x \notin D$. (by definition of $C \cap D$)

We can also prove it by $p \rightarrow q$

$$\forall x [(x \in A \wedge x \in B) \rightarrow (x \in C \wedge x \notin D)]$$

Assume $x \in A$

$$\forall x [\neg(x \in A \wedge x \in B) \vee (x \in C \wedge x \notin D)]$$

Assume $x \in A$ & $x \in D$

$$\forall x [\neg(T \wedge x \in B) \vee (x \in C \wedge F)]$$

$$\forall x [\neg(x \in B) \vee F]$$

$$\forall x (x \notin B)$$

10) Give: $a, b \in \mathbb{R}, a < b$
 Prove: $(a+b)/2 < b$

Assume $a < b$, adding b to both sides gives $a+b < 2b$

dividing by 2 $(a+b)/2 < b$. (we did not prove that we can change the sign of inequality)

11) Give: $x \in \mathbb{R}, x \neq 0, (\sqrt[3]{x} + 5) / (x^2 + 6) = 1/x$

Prove: $x \neq 8$

We will prove contrapositive. Suppose $x = 8$

$$(\sqrt[3]{8} + 5) / (8^2 + 6) = (2 + 5) / (64 + 6) = 7/70 = 1/10 \neq 1/8$$

Proved

12) Given: $a, b, c, d \in \mathbb{R}$, $0 < a < b$, $d > 0$

$$a < c, bd > 0$$

$$\text{Goal: } c > d \quad (\text{neg})$$

we will prove contrapositive. Suppose $c > d$
 Multiply by a gives $a < c \leq ad$ (a itve, so no equality
 Now $a < b$. Multiplying by d , we get $ad < bd$ (since))

$$\text{In other words } a < c \leq ad < bd \rightarrow a < bd$$

Proved.

($\neg p$)

$\neg q \rightarrow \neg p$

13) Given $x, y \in \mathbb{R}$, $3x + 2y \leq 5$, $x > 1$

$$\text{Goal: } y < 1$$

By contraposition let $y \geq 1$, then $2y \geq 2$

$$\text{Q. } 3x + 2y \geq 3x + 2 \leq 5$$

$$\text{Q. } 3x + 2 \leq 5 \quad \text{Q.}$$

$$\text{Subtracting 2, we get } 3x \leq 3 \rightarrow x \leq 1$$

Proved

14) $x, y \in \mathbb{R}$, $x^2 + y = -3$, $2x - y = 2$

$$\rightarrow x = -1$$

we will prove directly. Solving this system directly yield
 only one solution $\{-1, 1\}$. so $x = -1$ Proved.

15) $x > 3$ and $y < 2$

$\rightarrow x^2 + 2y > 5$

As $x > 3$, $x^2 > 9$ & $y < 2$ $-2y > -4$ ($\therefore$ multiplying by -2)
 Adding these two, $x^2 - 2y > 5$ (eqs & inequalities valid if we add or subtract them)

16) (a) This proof is of form $q \rightarrow p$ rather than $p \rightarrow q$. It says that if conclusion is True, then hypothesis is True as well, however it can be that hypothesis is True without conclusion being True.

(b) ~~Contrapositive~~ Suppose $x \neq 7$ suppose $2x - 5 / x - 4 = 3$ is T
 $2x - 5 = 3(x - 4) \rightarrow 2x - 5 = 3x - 12 \rightarrow x - 7 = 0$
 $\rightarrow x = 7$

17) (a) Also of form $q \rightarrow p$ rather than $p \rightarrow q$.

(b) $x = 0$ is a counterexample. $0^2 y = 9y$ $0y = 9y \rightarrow 0 = 0$
 (Here y need not be 0, y can be anything here)
 $x = -3$ is also counterexample. It gives $y = y$, so y can be anything not necessarily 0.
 The problem lies in assumption that $x^2 \neq 9$ from hypothesis $x \neq 3$ because there is -3 for which $x^2 = 9$ [$(-3)^2 = 9$].
 This assertion is wrong.